

EcoPrice: A Real-Time Dynamic Pricing and Spoilage Prediction Algorithm

Team

Manasa Santosh (23BCB7139)

M. Sri Harshitha (23BCB7177)

P. Alex Rohan (23BCB7225)

R. Anjana (23BCB7032)

Program

Computer Science and Business Systems (CSBS)

Date

August 2026

Abstract

This presentation introduces **EcoPrice**, a real-time dynamic pricing and spoilage prediction algorithm designed for the retail grocery sector. By integrating time-series forecasting (XGBoost) with reinforcement learning (PPO), EcoPrice autonomously adjusts perishable goods pricing to minimize physical food waste while maintaining revenue viability — without reliance on expensive item-level hardware tracking.

Agenda

1	Introduction Background, current limitations, and the algorithmic shift toward data-driven pricing
2	Literature Survey Tabular review of prior methods, advantages, and disadvantages
3	Problem Statement Two precisely scoped research problems motivating this work
4	Objectives Two targeted technical objectives aligned to the problem statement
5	Proposed System Architecture Placeholder — pending reviewer confirmation
6	References APA-formatted citations for all surveyed works

Introduction

The Perishable Inventory Challenge

High Waste Rates in Retail Grocery

The retail grocery sector suffers from high rates of physical waste due to the rigid pricing of perishable goods. Static price tags fail to respond to the biological reality of product degradation, resulting in unsold stock that must be discarded.

Strict Shelf Life Constraints

Items with strict shelf lives (**1 to 30 days**) require continuous monitoring to ensure they are sold before expiration. The narrow window between optimal sale and spoilage demands near-real-time decision-making that manual processes cannot sustain.

Scale and Complexity of Manual Management

Managing this inventory manually across hundreds of product categories is inefficient and error-prone. Human oversight cannot simultaneously track expiry dates, demand signals, and competitor pricing at the speed and granularity required for effective markdown decisions.

The Current Flaw

Limitations of Static Markdowns

The Traditional Approach

Retailers traditionally rely on **manual, threshold-based markdown strategies** — for example, applying 50% discount stickers at the end of the day. This approach is rule-based, time-triggered, and entirely uniform regardless of product condition or market context.

Disconnected from Real-Time Signals

These static strategies are entirely disconnected from real-time market demand, weather changes, and competitor pricing — three variables that directly influence consumer purchasing behavior for perishables.

Dual Failure Mode

Consequently, inventory either **spoils entirely** or is sold at an **unnecessarily deep discount**, damaging both sustainability and revenue. Neither outcome is acceptable in a margin-sensitive retail environment.

The Algorithmic Shift

Transitioning to Data-Driven Pricing

1

From Manual Guessing

Modern data science enables the transition from manual guessing to **autonomous, dynamic price adjustments** — removing human latency from the markdown decision loop entirely.

2

Time-Series + Reinforcement Learning

By combining **time-series forecasting** with **reinforcement learning**, pricing policies can be optimized continuously. The model learns from historical demand patterns and adapts to live inventory signals.

3

Demand-Aligned Clearance

This algorithmic approach dynamically aligns stock levels with consumer demand to **clear inventory exactly as it approaches its expiration date** — maximizing both revenue recovery and waste reduction simultaneously.

Literature Survey

S. No	Author Name	Method	Advantage	Disadvantage
1	Zhu et al. (2024)	Hybrid Prophet-XGBoost	High accuracy for predicting perishable product demand.	Purely predictive; does not prescribe specific pricing actions.
2	Yavuz and Kaya (2024)	Deep RL (DQN, PPO)	PPO shows stable convergence for joint pricing.	Confined entirely to digital simulation environments.
3	Ali et al. (2025)	Hierarchical Transformer + PPO	Reduces food waste over standard (s, S) inventory policies.	Optimizes primarily for profit; lacks explicit penalty for physical waste.
4	Wasteless Ltd. (US Patents)	Proprietary RL Engine	Directly connects dynamic pricing to Electronic Shelf Labels.	Relies heavily on expensive item-level RFID tracking.



All four surveyed works reveal a consistent gap: no existing method simultaneously enforces an explicit waste-minimization penalty *and* operates without item-level hardware dependency — the dual gap EcoPrice is designed to close.

Problem Statement

Problem 1 — Absence of Explicit Waste Penalty

Existing dynamic pricing algorithms optimize almost exclusively for profit in simulated environments, **lacking an explicit mathematical penalty to reduce actual physical food waste**. This omission means that spoilage is treated as an acceptable cost rather than a primary optimization target, resulting in algorithms that are commercially oriented but environmentally inadequate.

Problem 2 — Hardware Scalability Barrier

Current commercial architectures rely on **unscalable, expensive item-level hardware tracking (RFID)** which is financially unviable for standard retail integration. The capital expenditure required for full RFID deployment across a typical grocery store's SKU range renders these solutions inaccessible to the majority of retail operators.

Objectives

1

Develop an Integrated Predictive-Prescriptive Algorithm

To develop an integrated predictive-prescriptive algorithm (**XGBoost + PPO**) that autonomously adjusts perishable goods pricing to **explicitly minimize physical spoilage**. The reward function of the PPO agent will incorporate a direct mathematical penalty term for units that expire unsold, ensuring waste reduction is a first-class optimization objective alongside revenue.

2

Eliminate Item-Level Hardware Dependency

To eliminate the requirement for item-level hardware tracking by **predicting demand through a unified data preprocessing pipeline**. Demand signals will be inferred from aggregated transactional data, weather feeds, and temporal features — removing the need for RFID infrastructure and enabling deployment in standard retail environments without capital hardware investment.

Proposed System Architecture

System architecture diagram to be added once confirmed by the project reviewer. The placeholder below outlines the anticipated component structure pending final approval.

01

Data Ingestion Layer

Unified preprocessing pipeline ingesting transactional POS data, weather API feeds, competitor pricing signals, and product expiry metadata — without item-level RFID dependency.

02

Demand Forecasting Module (XGBoost)

Time-series demand forecasting using XGBoost trained on historical sales, seasonality features, and external covariates to generate short-horizon demand probability distributions per SKU.

03

Pricing Policy Engine (PPO Agent)

Proximal Policy Optimization (PPO) reinforcement learning agent that consumes demand forecasts and current inventory state to prescribe optimal markdown prices, with an explicit spoilage penalty term in the reward function.

04

Output / Integration Layer

Real-time price recommendations delivered to Electronic Shelf Label (ESL) systems or POS override interfaces for immediate in-store implementation.

 Architecture diagram is a placeholder — content is pending reviewer confirmation and must not be treated as final.

References

Ali, T., et al. (2025). Integrated Deep Learning and Reinforcement Learning for Perishable Food Supply Chains. *Proceedings of the 1st Engineering Data Analytics and Management Conference*, Atlantis Press.

Yavuz, M., & Kaya, O. (2024). Deep Reinforcement Learning for Joint Pricing and Inventory Management of Perishable Goods. *Applied Soft Computing*.

Wasteless Ltd. (2021). *System and method using a dynamic pricing engine to determine pricing for goods* (US Patent 11,182,811 B2). U.S. Patent and Trademark Office.

Zhu, Y., et al. (2024). Prediction of customer demand for perishable products using the hybrid Prophet-XGBoost model. *Applied Economics Letters*.